

Talmud-IR: A Talmud-Inspired Interface for Discussing RAG Response Quality

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Abstract. Retrieval-augmented generation (RAG) systems promise factually grounded answers, yet evaluating their quality remains difficult. Automated metrics and LLM-as-judge approaches offer scalability but risk circularity, benchmark leakage, and loss of diversity. Human assessors, meanwhile, often struggle to notice subtle omissions or hallucinations when responses appear linguistically fluent and confident. We present *Talmud-IR*, a novel user interface inspired by the dialogic structure of the Talmud. It visualizes RAG outputs as a central text surrounded by layers of evidence, commentary, and meta-assessment, enabling sustained human–LLM discussion about system quality and failure priorities. The prototype supports comparative RAG evaluation, collaborative exploration of “unknown unknowns,” and pedagogical use for teaching critical reading of AI-generated content.⁷

Keywords: RAG · LLM judge · Exploratory Evaluation.

1 Introduction and Motivation

Retrieval-augmented generation (RAG) systems integrate document retrieval with large language models (LLMs) to produce grounded, conversational answers. Despite rapid progress, evaluating RAG outputs remains challenging. Automatic evaluation with LLM judges is convenient but introduces risks such as circularity, test-set leakage, and overfitting to benchmarks [5]. Human judgments remain vital, yet even expert assessors can overlook subtle factual errors or missing context when presented with fluent, confident prose [2, 8].

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⁷ Code: <https://github.com/WojciechKusa/talmud-ir>

Prototype: <https://wojciechkusa.github.io/talmud-ir/>

Traditional IR evaluation treats judgment as a cost trade-off: How much assessor time can we buy, and how many topics can be covered? For generative systems, however, quality cannot be captured by one-off relevance labels. Responses may differ in nuance, perspective, and completeness, and a single evaluation paradigm cannot easily capture their multifaceted quality. To reveal such “*unknown unknowns*”, we argue for exploratory evaluation interfaces that foster sustained dialogue between humans and machines.

We draw inspiration from the Talmudic tradition, which has supported centuries of evolving discourse around a shared text [10]. The Talmud’s structure—a central law text (the *Mishnah*) surrounded by interpretative discussions (the *Gemara*) and later super-commentaries—offers a compelling analogy for exploratory discussions about quality of AI-generated content. The Jewish tradition of Talmudic scriptural interpretation is based precisely on not setting any interpretation as absolute, but rather on attributing intrinsic value to the discourse surrounding different interpretations [1]. Ideas that have enabled critical conversation to persist over millennia may also help us build lasting frameworks for discussing RAG output quality criteria.

2 Related Work

A growing body of research explores how to evaluate the quality of RAG outputs. Prompt-based approaches, such as UMBRELA [18], LLM-as-a-judge [20], and RAGAS [6] use large language models as judges, relying on direct prompting to assign quality labels and judge RAG outputs. While attractive for their low cost and scalability, these methods are inherently fragile: they risk circularity when the same model family serves as both generator and judge [3], and are susceptible to test-set leakage and benchmark memorization [4, 5], and often conflate confident phrasing with factual adequacy. As a result, such evaluations can obscure the very failure modes they aim to detect.

To mitigate these issues, hybrid judging [15], nugget- or rubric-based methods [7, 11, 14, 16] offer a middle ground between human and automatic assessment. Here, evaluators specify which facts or ideas (*nuggets*) should appear in a high-quality answer, while LLMs assist in aligning those nuggets to generated text. Although this improves transparency, coverage of the “right” nuggets alone does not guarantee a good response: information must also be organized logically, expressed coherently, and contextualized with sufficient background. Understanding such aspects of quality requires richer, discourse-oriented evaluation artifacts.

Another recent direction concerns the *faithfulness of citations* in RAG outputs. Systems such as ARGUE and AutoArgue [12, 19] verify whether retrieved passages genuinely support the generated statements, linking factual grounding to user trust. Yet, even when coverage and citation accuracy are accounted for, we remain uncertain about what the *next most critical failure point* of RAG systems will be—whether it involves reasoning, context management, or the presentation of evidence. To uncover these evolving weaknesses quickly, we need

interfaces that allow humans and LLMs to collaboratively explore, debate, and surface new evaluation priorities as they emerge.

3 Prototype: Talmud-Style Interface for RAG Evaluation

The *Talmud-IR* prototype adopts the layout of a Talmudic page (Figure 1), transforming evaluation into a structured, multi-participant conversation. The interface centers on a query–response pair produced by a RAG system and surrounds it with layers of supporting information and commentary. The central UI elements are:

- **Central text (Mishnah):** the user query and the system’s generated answer.
- **Inner margins:** retrieved snippets that provide evidential grounding, highlighting faithful or contradictory citations.
- **Outer margins (Gemara):** threaded discussions among human experts and LLM-based assessors reflecting on correctness, completeness, tone, or omissions. Comments can reference specific spans of text or earlier remarks, forming a living discourse.
- **Super-commentaries:** a meta-layer aggregating evaluation metrics such as nugget coverage or citation validity, summarizing consensus across assessors.

The prototype is implemented as a lightweight web application built with React 19 and Tailwind CSS. It links textual spans to snippets and commentaries. Selecting a phrase in the central response highlights corresponding evidence; hovering over snippets reveals linked commentary. A sidebar records earlier discussions so recurring issues can be revisited, mirroring how later scholars build upon prior interpretations.

The typical workflow of the prototype user begins with inspecting the query and the RAG response at the center. During this inspection, the provided evidence can be used to validate aspects such as factuality. By leaving comments on specific aspects and spans, the prototype user enriches the discussion around the response’s quality. Comments from previous human users can be reviewed and reacted to. Additionally, commentary from LLM-based systems for RAG evaluation is shown and can be considered during human inspection. This helps the prototype user conduct an in-depth analysis of the given RAG response.

An alternative configuration of the prototype system can invert the hierarchy of RAG responses and supporting documents: a cited document may appear at the center, surrounded by multiple RAG responses that interpret or misinterpret it. This view supports comparative studies of hallucination and evidence selection.

Our implementation can integrate with nugget-based evaluation systems [7, 12, 14] to visualize nugget-based metrics alongside qualitative commentary. By unifying interpretative discussion with quantitative measures, the system re-frames evaluation as an ongoing, collaborative conversation.

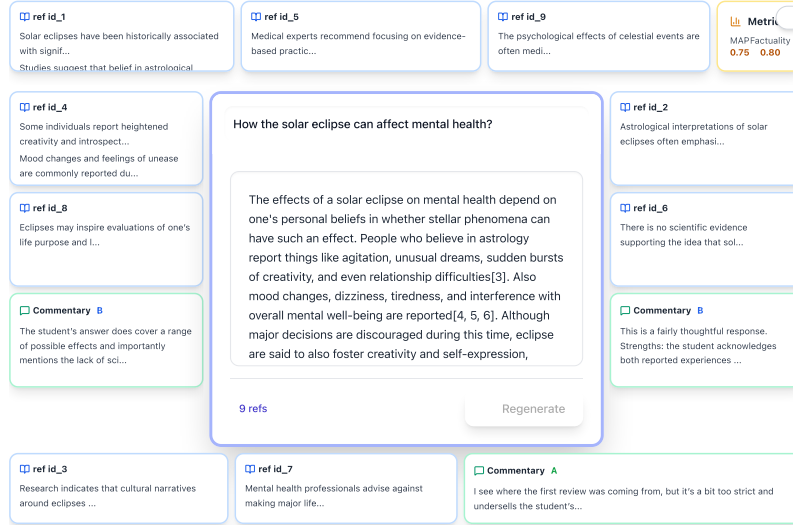


Fig. 1: Talmud-IR prototype: central RAG response (*Mishnah*) surrounded by retrieved evidence and human/LLM commentary (*Gemara*). While inspecting the given RAG response, the prototype user can take into account the given additional evidence and commentary. The example presented in this figure shows a RAG response from the *webis-manual* submission [9] for the TREC 2024 RAG Track [13, 17].

While Figure 1 shows a particular example from the TREC 2024 RAG Track, the prototype supports any RAG run, including responses and external evaluation metadata, as long as it uses our input format. This makes it applicable to a wide range of RAG systems and tasks.

4 Intended User Groups

The Talmud-IR interface serves multiple stakeholder communities involved in the evaluation and use of RAG systems. **Human assessors** who compare multiple system outputs benefit from the structured presentation of retrieved evidence and the opportunity to anchor their judgments within an evolving commentary, improving both transparency and consistency. **Evaluation developers** can analyze discussion threads to study agreement patterns, biases, and hybrid human–LLM assessment workflows. In professional domains such as law, medicine, or scientific retrieval, **domain experts** can trace how generated statements are grounded in cited documents, supporting more trustworthy decision-making. Finally, **educators and learners** can use the interface as a pedagogical tool to

explore how AI systems reason, argue, and fail—turning evaluation discourse into an exercise in critical reading and collaborative sense-making.

5 Discussion and Outlook

Our Talmud-IR prototype reimagines evaluation as a dialogic rather than a purely quantitative process. By capturing interpretative layers around each response, it allows researchers to document emerging quality criteria (e.g., plausibility, coverage, fairness) and track which criteria have been addressed or newly arise as systems improve. Future extensions will incorporate analytics to detect convergence among assessors and integrate LLM summarization of discussions. We plan to use Talmud-IR in collaborative workshops (e.g., TREC or CLEF tracks) as a means to guide the discussion about quality.

We invite the ECIR community to engage with the demonstration and to consider how dialogic interfaces like Talmud-IR can complement automated metrics by fostering a sustained, evidence-driven conversation about the quality of RAG systems.

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